Binary Search

Topic: DivideAndConquer File: d3.pdf

PROBLEM

Find a target value in a sorted array, returning its index or -1 if missing.

DIVIDE

Inspect the middle element and discard the half that cannot contain the target.

PSEUDOCODE

```
low = 0; high = n - 1
while low <= high:
    mid = low + (high - low) // 2
    if A[mid] == x: return mid
    if A[mid] < x: low = mid + 1
    else: high = mid - 1
return -1
```

EXAMPLE

A = [1, 4, 6, 9, 12, 20], target 12. Each comparison halves the remaining search range.

COMPLEXITY

Time: O(log n). Iterative extra space: O(1). The input must be sorted.